TwoWin-SOVA: Two Windows Discrete Cosine Transform and Synthetic Minority Oversampling Technique One-Versus-All Ensemble Classifiers for Imbalanced Hyperspectral Image Explainable Classification

Ziqi Zhao, Changbao Yang, and Qiong Wu

***Abstract*— Patchwise methods have been widely used in hyperspectral image (HSI) classification. In HSI classification methods that use only spectral information, a large number of unlabeled pixels, known as background pixels, are removed before training, which loses spatial information. Patchwise methods obtain patches through a local window to retain the spectral–spatial information in background pixels. Larger patches have more background pixels, but more redundant information in the extracted high-dimensional features reduces classification accuracy. Due to the complex and uneven distribution of land cover classes, class imbalance is a common problem in HSI. In this article, a two windows discrete cosine transform and synthetic minority oversampling technique one-versus-all ensemble classifiers (TwoWin-SOVA) method is proposed for addressing the problems of high dimensionality caused by large patches and class imbalance, which consists of feature extraction and ensemble classifiers. A two windows discrete cosine transform (TwoWin-DCT) is utilized for feature extraction, compressing large patches to extract spectral–spatial features, which achieves dimensionality reduction. Index features, which represent the correlation between spectral bands, are fused with them as spectral–spatial-index features. Synthetic minority oversampling technique one-versus-all ensemble classifiers (SOVA) construct ensemble classifiers utilizing one-versus-all (OVA) strategy with synthetic minority oversampling technique (SMOTE) and the base classifiers of light gradient boosting machine (LightGBM) to address class imbalance problem. The experimental results on the five public HSI datasets show that TwoWin-SOVA effectively tackles the problems of high dimensionality caused by large patches and class imbalance, achieving classification performance as good as several state-of-the-art methods.**

***Index Terms*— Class imbalance, ensemble learning, explainable, feature extraction, hyperspectral image (HSI) classification, patch.**

I. Introduction

HYPERSPECTRAL image (HSI) presents a 3-D cube with spatial information in two dimensions and abundant spectral information in the third one consisting of hundreds of spectral bands, widely used in land cover analysis, precision agriculture, environmental monitoring, and urban planning [1], [2]. HSI classification labeled pixels to identify land cover classes of interest, and the high volume of spectral bands aids in discriminating different materials [3]. The HSI datasets consist of limited labeled pixels and a large number of unlabeled background pixels containing abundant spatial information [4]. Assuming that the pixels of the same class may be spatially adjacent, the current patchwise methods extract spatial information through a local window and discriminate different objects with similar spectral characteristics by shapes and textures [5], [6]. Larger patches contain more spatial information and can provide more spectral–spatial features. Curse of dimensionality, known as Hughes phenomenon, may emerge with higher dimensional features, that is to say, the accuracy of the classifier learned from limited training samples first rises and then declines with increasing dimensionality [7], [8]. With increasing patch size, the classification accuracy usually tends to rise and then decline [9], [10], [11]. Excessive redundant information in the large patches limits the patch size and affects the classification performance of methods using large patches, such as the patch size in [9] reaching the optimal performance at 27×27 and the patch size in [10] reaching the optimal performance at 21×21. Feature selection selects a subset of features to optimize classifier performance [12]. Discrete cosine transform (DCT) compresses image energy in low-frequency coefficients, and retaining low-frequency coefficients enables feature selection for dimensionality reduction [13]. The emergence of feature selection discards redundant information in high-dimensional features, allowing spectral–spatial features to be extracted from larger patches, which provides more spatial information to the classifier.

The class imbalance problem in HSI increases the learning difficulty of a classifier, and the deterioration of classifier performance by class imbalance heavily depends on data properties and the selection of classifiers [14]. There are currently three categories of approaches to deal with class imbalance: 1) data-level approaches resample to balance the class distribution; 2) algorithm-level approaches adapt classi-

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fication algorithms to bias learning toward minority classes; and 3) cost-sensitive learning frameworks define misclassification costs during classifier learning, which are not available before learning. Ensemble learning constructs an ensemble of classifiers, aggregates the predictions of several base classifiers to improve classification performance and generalization ability, and is also used to deal with class imbalance problem [15], [16]. The class imbalance in HSI is usually multiclass imbalance due to numerous land cover classes. The ensemble learning strategies of one-versus-one (OVO) and one-versus-all (OVA) are usually used to address multiclass imbalance, transforming a multiclass imbalance problem into multiple binary problems, OVO constructs classifiers for all combinations between pairs of classes, and OVA learns classifiers for each class and a combination of all other classes [17]. An OVA ensemble has fewer base classifiers, less runtime, and a simpler framework than an OVO.

Appropriate spectral–spatial fusion methods can significantly improve HSI classification [2]. Deep learning-based methods with automatic feature learning have demonstrated the most superior classification performance in HSI classification [18], [19], [20], [21]. Convolutional neural network (CNN)-based methods automatically extract spectral–spatial features for end-to-end classification [22], [23]. The embedding of the attention mechanism allows the network to capture salient regions, enhancing the representation capacity of features [24], [25]. Assuming that the pixels of the same class are segmented into an object in the scene, superpixel-based methods use superpixels to extract spectral–spatial features for training a classifier [26], [27]. Sparse representation-based methods represent each test sample as a sparse linear combination of all training samples to achieve HSI classification [28]. The fast patch-free global learning (FPGA) framework was proposed to address the limitation of the existing patchwise methods, but limited in HSI datasets with insufficient and imbalanced samples [5], [29]. Patchwise methods extract spatial features with a window of fixed or adaptive size, stack extracted spectral features and spatial features, or jointly extract spectral–spatial features to generate a spectral–spatial cube.

This article proposes a two windows DCT and synthetic minority oversampling technique OVA ensemble classifiers (TwoWin-SOVA) method for addressing the problems of high dimensionality caused by large patches and class imbalance in HSI classification, and this method mainly consists of two windows DCT (TwoWin-DCT) module and synthetic minority oversampling technique OVA ensemble classifiers (SOVA) module. The TwoWin-DCT is utilized to extract the spectral–spatial features of large patches, in which the DCT window is utilized for dimensionality reduction, and index features represent the correlation between spectral bands as supplements to spectral–spatial information. SOVA is utilized to deal with the multiclass imbalance problem in HSI. The main contributions of this article are as follows.

1) TwoWin-DCT is proposed as a feature extraction method to reduce high dimensionality caused by large patches. This method obtains large patches in HSI through a local window as the inputs of DCT. A triangular DCT window is utilized for feature selection, and only the low-frequency coefficients of the patches after DCT compression are retained to achieve dimensionality reduction. The triangular patches are flattened into vectors and stacked as spectral–spatial features.
2) Index features supplement the spectral–spatial information to identify land cover classes with similar spectral curves but the different correlation between spectral bands. Normalized difference vegetation index (NDVI), normalized difference moisture index (NDMI), normalized difference water index (NDWI), green-red vegetation index (GRVI), green NDVI (GNDVI), and ironoxide (IO) are the index features. The spectral–spatial features and the index features are fused as spectral–spatial-index (SSI) features.
3) SOVA constructs ensemble classifiers by the OVA strategy to address the class imbalance problem in HSI. OVA transforms a complex multiclass imbalance problem into several simpler binary imbalance problems. Synthetic minority oversampling technique (SMOTE) oversamples the minority class in each binary imbalanced training set. Light gradient boosting machine (LightGBM) can learn each base classifier fast with the high-dimensional SSI features.
4) Shapley additive explanations (SHAP) is a machine learning explainable tool for explaining the contributions of the SSI features to predictions by calculating Shapley values, and its visualization plots can explain that the extracted features aid the SOVA to achieve great classification performance.

The remainder of this article is organized as follows. Section II introduces the problems and solutions. Section III describes the TwoWin-SOVA method. Section IV analyzes the experimental setups and results. Section V discusses the prediction results. The conclusions are drawn in Section VI.

II. Background

In HSI classification, the problems of high dimensionality caused by large patches and class imbalance affect classification performance. In this article, the problem of high dimensionality caused by large patches is addressed during the design of the feature extraction module and the problem of class imbalance is addressed during the design of the ensemble classifiers module.

A. Problem of High Dimensionality Caused by Large Patches

HSI has strong spatial homogeneity, in which land covers usually exist in patches of different sizes. Pixels in a patch belong to the same land cover class, so the pixels of the same class may be spatially adjacent. The land cover classes in HSI are difficult to distinguish by visual interpretation, and the process of labeling huge numbers of pixels is time-consuming and laborious, causing limited labeled pixels and a large number of unlabeled background pixels [4]. The background pixels are usually removed before the classifier learning, which causes a large loss of spatial information and makes it difficult to

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discriminate the pixels with similar phenological and spectral characteristics. Patchwise methods obtain patches through a local window, flatten patches into vectors, and stack them as the spectral–spatial features [29]. Larger patches with more background pixels contain more spatial information but generate high dimensionality and massive redundant information, which may result in Hughes phenomenon.

B. Feature Extraction

High-dimensional space is mostly empty, which implies that high-dimensional features can be projected into a lower dimensional subspace, without losing class separability information [30]. DCT can be interpreted as clustering the sum of all pixel intensities in the low-frequency coefficients which is the upper left corner of the patch after DCT, and removing the remaining high-frequency coefficients can achieve dimensionality reduction [13]. In this article, DCT is used as a method of feature extraction to compress most of the information in the high-dimensional features into a subset and select this subset by retaining the low-frequency coefficients, which achieves feature selection. Due to the energy compaction property, DCT is universally utilized for lossy data compression of signals and images [31], [32], [33]. For a given patch matrix $\mathrm{Mat} \in \Re^{m\times n}$, DCT is defined as

$$\mathrm{DCT}(i,j) = a_i a_j \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} \mathrm{Mat}(m,n) \times \cos\frac{\pi(2m+1)i}{2M} \times \cos\frac{\pi(2n+1)j}{2N} \quad (1)$$

$$0 \le i \le M, \quad 0 \le j \le N \quad (2)$$

where

$$a_i = \begin{cases} \dfrac{1}{\sqrt{M}}, & i=0 \\ \sqrt{\dfrac{2}{M}}, & 1 \le i \le M-1 \end{cases} \quad (3)$$

$$a_j = \begin{cases} \dfrac{1}{\sqrt{N}}, & j=0 \\ \sqrt{\dfrac{2}{N}}, & 1 \le j \le N-1. \end{cases} \quad (4)$$

Some land cover classes in HSI have similar spectral curves but the different correlation between spectral bands [5]. The various indices represent spectral transformations between spectral bands sensitive to different land cover characteristics and are more useful indicators of state and condition than individual spectral band [34]. The phenological indices calculated in this article are NDVI, NDMI, NDWI, GRVI, and GNDVI, and the geological index is IO. NDVI is sensitive to chlorophyll but saturated with flourish vegetation [35]. NDMI is sensitive to the moisture levels in vegetation [36]. NDWI is sensitive to surface water [37]. GRVI is sensitive to land cover types [38]. GNDVI is sensitive to photosynthetic activity [39]. IO is sensitive to iron oxides [40]. The above indices are calculated as follows:

$$\mathrm{NDVI} = \frac{\mathrm{NIR}-\mathrm{Red}}{\mathrm{NIR}+\mathrm{Red}} \quad (5)$$

$$\mathrm{NDMI} = \frac{\mathrm{NIR}-\mathrm{SWIR}}{\mathrm{NIR}+\mathrm{SWIR}} \quad (6)$$

$$\mathrm{NDWI} = \frac{\mathrm{Green}-\mathrm{NIR}}{\mathrm{Green}+\mathrm{NIR}} \quad (7)$$

$$\mathrm{GRVI} = \frac{\mathrm{Green}-\mathrm{Red}}{\mathrm{Green}+\mathrm{Red}} \quad (8)$$

$$\mathrm{GNDVI} = \frac{\mathrm{NIR}-\mathrm{Green}}{\mathrm{NIR}+\mathrm{Green}} \quad (9)$$

$$\mathrm{IO} = \frac{\mathrm{Red}}{\mathrm{Blue}} \quad (10)$$

where blue, green, red, near-infrared (NIR), and short-wave infrared (SWIR) are the pixel intensity matrices of the corresponding spectral bands of HSI, respectively.

C. Problem of Class Imbalance

Class imbalance is prevalent in real-world applications and is an inherent limitation of datasets [41]. A classifier tries to maximize its prediction accuracy, and the classifier affected by the class imbalance problem has high accuracy overall but poor performance on the minority class. Current methods address class imbalance by adapting algorithms, resampling, defining misclassification costs, and ensemble learning. Resampling aims to correct the distribution problem of a training set, and the optimal distribution for training a classifier is sometimes not the distribution of the original dataset [42]. Ensemble learning trains several classifiers and inducts the predictions to improve the accuracy of an ensemble and usually combined with other methods. The advantage of the hybrid methods is that only the ensemble learning algorithm needs to be modified without changing the base classifier. Combining resampling and ensemble learning can address the challenge of achieving desired classification accuracy on the data with a skewed class distribution.

D. Ensemble Classifiers

Ensemble learning inducts and combines several classifiers to improve the classification performance of the ensemble and has a synergy with resampling. The proposed method combines OVA and SMOTE to construct ensemble classifiers. OVA constructs a binarization framework with a divide and conquer strategy and uses several binary classifiers to undertake the multiclassification [17]. OVA divides an m-class imbalanced dataset into m binary imbalanced datasets by combining each class and the others and uses each binary imbalanced dataset to learn a base classifier. The output of the OVA ensemble is a result vector R of length m

$$R = (r_1, r_2, \ldots, r_i, \ldots, r_m) \quad (11)$$

where r_i is the prediction value of the ith base classifier, and the class corresponding to the maximum prediction value is the predicted class of the input test sample. Before training a base classifier, SMOTE is utilized to oversample the minority class and generates a binary balanced dataset. SMOTE generates new samples between adjacent minority samples by interpolation and has been applied in several fields due to its simplicity and robustness [43], [44], [45]. SMOTE provides more training samples for the classifier and mitigates the insufficient sample

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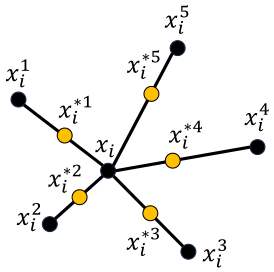


Fig. 1. Illustration of SMOTE.

problem in HSI. For a given minority sample x_i, SMOTE is defined as

$$x_i^{*p} = x_i + u\left(x_i^p - x_i\right). \tag{12}$$

where x_i^{*p} is a synthetic minority sample, u is a random number between 0 and 1, and x_i^p are the K randomly selected samples from the neighborhood, $p = 1, \ldots, K$. Taking $K = 5$ as an example, the SMOTE is illustrated in Fig. 1.

LightGBM is a new implementation of gradient boosting decision tree (GBDT), which is modeled as the base classifier of the OVA ensemble classifiers in this article [46]. For large sample data and high-dimensional features, LightGBM proposed gradient-based one-side sampling (GOSS) and exclusive feature bundling (EFB) to obtain satisfactory efficiency and scalability. The GOSS reduces data size by retaining large gradient samples and randomly dropping small gradient samples. The EFS reduces the number of features by bundling mutually exclusive features in a high-dimensional sparse feature space into a single feature. The above two new algorithms allow LightGBM to train over 20 times faster than GBDT without hurting accuracy. For the high dimensions of the SSI features, this article utilizes LightGBM to rapidly train base classifiers with high accuracy, and the input of a base classifier is a balanced binary training set after SMOTE.

III. TwoWin-SOVA: Two Windows Discrete Cosine Transform and Synthetic Minority Oversampling Technique One-Versus-All Ensemble Classifiers

TwoWin-SOVA is a patchwise method for pixelwise supervised classification, which consists of two modules: feature extraction and ensemble classifiers. In feature extraction, TwoWin-DCT obtains spectral–spatial features and fuses them with the index features as SSI features. In ensemble classifiers, SOVA learns each base classifier and constructs the ensemble classifiers rapidly. The flowchart of the proposed method is shown in Fig. 2.

A. TwoWin-DCT

Similar to the feature extraction of most patchwise methods, the TwoWin-DCT obtains patches to represent spatial information through a local window, the difference is that DCT is used to cluster the pixel intensities in the upper left corner of the patch. Only the small triangular patch after DCT is retained allowing the TwoWin-DCT to obtain a larger patch containing more spatial information. The process of the TwoWin-DCT is shown in Fig. 3. Considering the dual window nested structure and reduced parameters of the TwoWin-DCT, the shape of the local window is a square with a side length w, and the shape of the DCT window is an equilateral right triangle with right-angled side length t, and w and t are parameters determined through experiments. The local window obtains patches centered on each pixel by a $w \times w$ window, assigning the spatial information of the neighborhood background pixels as that of the center pixel. For an HSI cube, a local window obtains patch cubes of each pixel with spectral–spatial information.

The TwoWin-DCT aims to cluster spectral–spatial information in a small structure of the patch cube. The large patches in the patch cube are the inputs to DCT, which cluster the spatial information in the low-frequency coefficients. The DCT window is utilized for feature selection, retaining the low-frequency coefficients to obtain the triangular patches. As shown in Fig. 3, the triangular patches in the patch cube are retained by the DCT window, flattened into vectors, and stacked to obtain the spectral–spatial features. In this study, the side length w of the local window is selected as 37 through experiments, and the side length t of the DCT window takes values $[1, ((w+1)/2)]$, which is selected as 5 through experiments. The dimensions of the spectral–spatial features are $((t^2+t) \times b/2)$, where b is the number of spectral bands. The TwoWin-DCT uses dual windows to compress spatial information in the larger patches with more background pixels into the small triangular patches, reducing the high dimensions of the spectral–spatial features.

An index is calculated from two or more spectral bands and represents phenological or geological characteristics. In this article, indices are used as supplements to spectral–spatial features for identifying land cover classes with the different correlations between spectral bands. The spectral band matrices corresponding to blue, red, green, NIR, and SWIR are selected based on sensor wavelength information for the index calculation. As the index features of the proposed method, each of NDVI, NDMI, NDWI, GRVI, GNDVI, and IO can be calculated by inputting the pixel intensity matrices, and the six index matrices are of the same size as the input matrices. The index features of each pixel are the values of the corresponding position in the index matrices, and the index feature of each pixel is a feature vector of length 6. The index feature vector is fused with the spectral–spatial feature vector into SSI feature vector by feature stacking, that is, concatenation, as the input of the ensemble classifiers.

B. SOVA

SOVA employs the ensemble learning scheme of OVA to divide a multiclass imbalance problem into several binary imbalance problems with a divide-and-conquer strategy and utilizes SMOTE and base classifiers of LightGBM to tackle each binary imbalanced training set, and the flowchart is shown in Fig. 4. As a binarization framework, OVA learns a base classifier for each class, dividing the multiclass distribution of a training set into a binary distribution of that class and the other classes. The base classifier simply discriminates the class from the others and outputs the predicted class of

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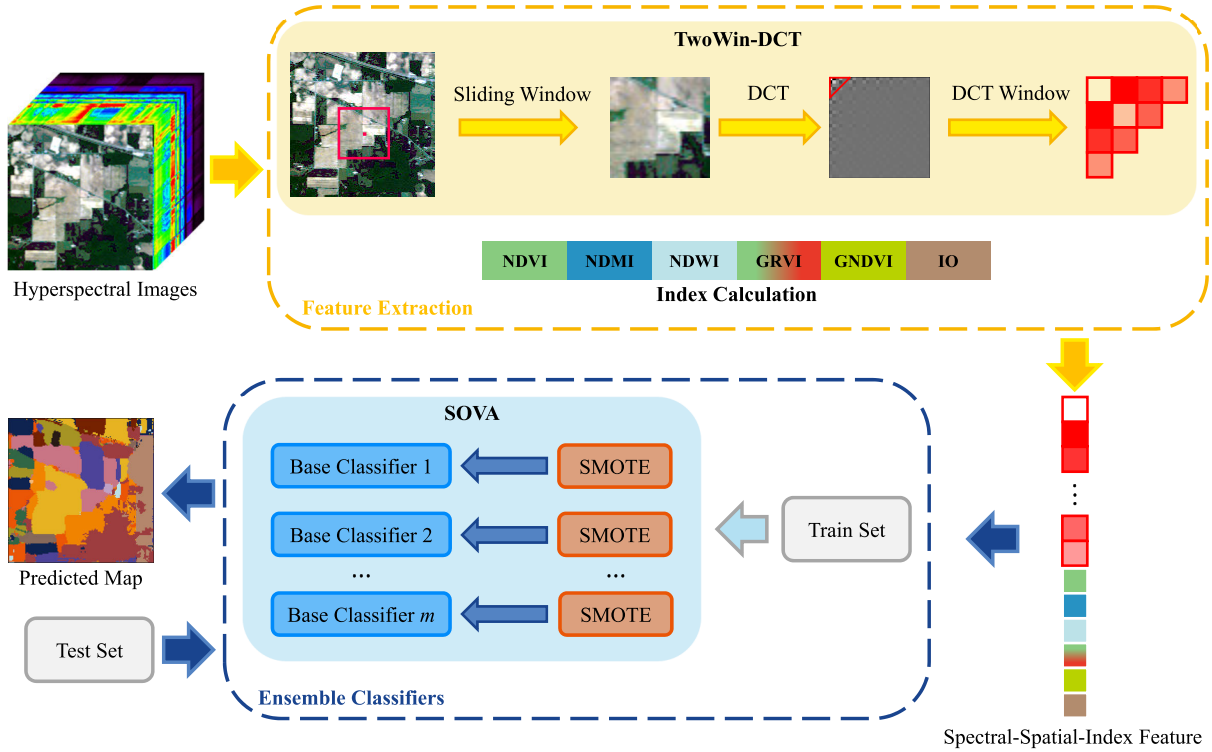


Fig. 2. Flowchart of the TwoWin-SOVA.

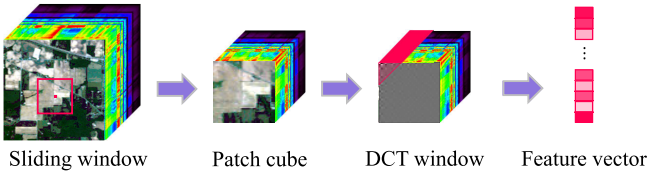


Fig. 3. Flowchart of the TwoWin-DCT.

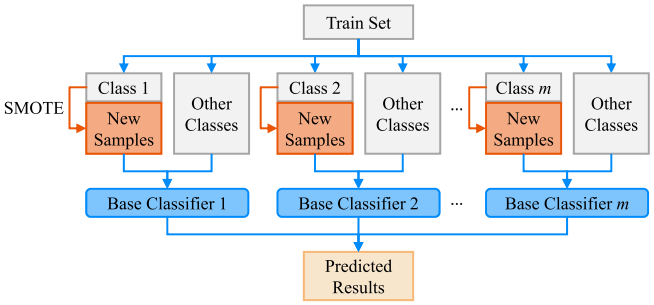


Fig. 4. Flowchart of the SOVA.

the input test sample. Compared with OVO which divides a training set into subsets with only two classes of samples, OVA maintains the data distribution of the original training set. The training set divided by OVA is usually binary imbalanced, and SMOTE is introduced to address the binary imbalance problem.

To ensure the quality of synthesized new samples, the proposed method utilizes a DCT window for feature selection before applying SMOTE, avoiding that a small number of samples in the high-dimensional feature space become most of the observed nearest neighbors [47]. SMOTE is based on the values of the features and their relationships, interpolates between the adjacent minority samples in the feature space, and then generates a balanced training set to overcome the adverse effects of the imbalanced dataset for training the classifier. Compared with undersampling, SMOTE avoids information loss and the insufficient sample problem without dropping any training samples that may be useful for classifier learning. Random oversampling (ROS) expands minority samples by simple replication, which is equivalent to weighting the minority class. SMOTE generates new samples to avoid weighting existing samples, overcomes the overfitting caused by simply replicating samples, and improves the generalization ability of the classifier.

LightGBM learns a high-accuracy base classifier rapidly by a balanced training set with high feature dimensions. Tree models have requirements for training sample size and feature quality. This article uses the training samples expanded by SMOTE and the SSI features, which represent spectral–spatial information and the correlation between spectral bands in HSI to aid LightGBM train high-performance classifiers. Each base classifier in SOVA predicts the test samples consistent with the natural class distribution by using the base classifiers learned from the training samples of the balanced distribution. SOVA combines oversampling and ensemble learning to address the class imbalance problem in HSI.

IV. Experimental Results

This article evaluates the classification performance of the proposed method on five public HSI datasets compared with several state-of-the-art methods. The degree of imbalance in

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES FOR THE INDIAN PINES DATASET

No.	Class	Train	Test	Total
1	Alfalfa	5	41	46
2	Corn-notill	143	1285	1428
3	Corn-mintill	83	747	830
4	Corn	24	213	237
5	Grass-pasture	48	435	483
6	Grass-trees	73	657	730
7	Grass-pasture-mowed	3	25	28
8	Hay-windrowed	48	430	478
9	Oats	2	18	20
10	Soybean-notill	97	875	972
11	Soybean-mintill	245	2210	2455
12	Soybean-clean	59	534	593
13	Wheat	20	185	205
14	Woods	126	1139	1265
15	Buildings-Grass-Trees-Drives	39	347	386
16	Stone-Steel-Towers	9	84	93
	Total	1024	9225	10249

the dataset is represented by the class imbalance ratio (IR), which is obtained by dividing the maximum sample size in each class by the minimum sample size. On the five datasets, the proposed method compares the classification performance with eight deep learning-based methods, including ContextualNet [20], SSAN [23], SSAtt [22], RSSAN [25], A^2S^2K-ResNet [24], SSSAN [10], SSTN [21], and CVSSN [11]. The above methods cover current representative deep learning patterns of 2-D CNN-based pattern, 3-D CNN-based pattern, spectral–spatial dual-branch pattern, two-subnetwork structure, residual learning pattern, and various attention mechanisms. Among them, SSSAN takes 21 × 21 patches as input and is also represented as the state-of-the-art method using large patches for comparison.

A. Datasets

The Indian Pines dataset is a 145 × 145 image of the Indian Pines test site in North-western Indiana with a spatial resolution of 20 m. The sensor is the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) with the wavelength range of 0.4–2.5 μm and 200 bands after removing the water absorption bands. The available ground truth contains 16 classes of land covers with 10 249 labeled pixels and 10 776 unlabeled background pixels. The IR of the dataset is 122.75, and the number of training and test samples for each class is shown in Table I.

The Pavia University dataset is a 610 × 340 image of the University of Pavia in northern Italy with a spatial resolution of 1.3 m. The sensor is the Reflective Optics System Imaging Spectrometer (ROSIS) with the wavelength range of 0.43–0.86 μm and 103 bands after removing the noise and water absorption bands. The available ground truth contains nine classes of land covers with 42 776 labeled pixels and 164 624 unlabeled background pixels. The IR of the dataset is 19.69, and the number of training and test samples for each class is shown in Table II.

TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR THE PAVIA UNIVERSITY DATASET

No.	Class	Train	Test	Total
1	Asphalt	663	5968	6631
2	Meadows	1865	16784	18649
3	Gravel	210	1889	2099
4	Trees	306	2758	3064
5	Painted metal sheets	134	1211	1345
6	Bare Soil	503	4526	5029
7	Bitumen	133	1197	1330
8	Self-Blocking Bricks	368	3314	3682
9	Shadows	95	852	947
	Total	4277	38499	42776

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES FOR THE KSC DATASET

No.	Class	Train	Test	Total
1	Scrub	76	685	761
2	Willow swamp	24	219	243
3	Cabbage palm hammock	26	230	256
4	Cabbage palm/oak hammock	25	227	252
5	Slash pine	16	145	161
6	Oak/broadleaf hammock	23	206	229
7	Hardwood Swamp	11	94	105
8	Graminoid marsh	43	388	431
9	Spartina marsh	52	468	520
10	Cattail marsh	40	364	404
11	Salt marsh	42	377	419
12	Mud flats	50	453	503
13	Water	93	834	927
	Total	521	4690	5211

The Kennedy Space Center (KSC) dataset is a 512 × 614 image of the KSC in Florida with a spatial resolution of 18 m. The sensor is the AVIRIS with the wavelength range of 0.4–2.5 μm and 176 bands after removing the water absorption bands. The available ground truth contains 13 classes of land covers with 5211 labeled pixels and 309 157 unlabeled background pixels. The IR of the dataset is 8.83, and the number of training and test samples for each class is shown in Table III.

The Houston dataset is a 349 × 1905 image of the University of Houston campus and its neighboring urban area with a spatial resolution of 2.5 m. The sensor is the ITRES-CASI (Compact Airborne Spectrographic Imager) 1500 with the wavelength range of 0.36–1.05 μm and 144 bands. The available ground truth contains 15 classes of land covers with 15 029 labeled pixels and 649 816 unlabeled background pixels. The IR of the dataset is 3.90, and the number of training and test samples for each class is shown in Table IV.

The HyRANK dataset contains two images of Dioni and Loukia, Dioni is a 250 × 1376 image and Loukia is a 249 × 945 image, both with a spatial resolution of 30 m. The sensor is the Hyperion with the wavelength range of 0.4–2.5 μm and 176 bands. The available ground truth contains 12 classes of land covers with 20 024 labeled pixels and 323 976 unlabeled background pixels in Dioni and 12 208 labeled pixels and 223

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TABLE IV
NUMBER OF TRAINING AND TEST SAMPLES FOR THE HOUSTON DATASET

No.	Class	Train	Test	Total
1	Healthy grass	125	1126	1251
2	Stressed grass	125	1129	1254
3	Synthetic grass	70	627	697
4	Trees	124	1120	1244
5	Soil	124	1118	1242
6	Water	33	292	325
7	Residential	127	1141	1268
8	Commercial	124	1120	1244
9	Road	125	1127	1252
10	Highway	123	1104	1227
11	Railway	123	1112	1235
12	Parking Lot1	123	1110	1233
13	Parking Lot2	47	422	469
14	Tennis court	43	385	428
15	Running track	66	594	660
	Total	1502	13527	15029

TABLE V
NUMBER OF EACH CLASS FOR THE HYRANK DATASET

No.	Class	Dioni	Loukia
1	Dense Urban Fabric	1262	288
2	Mineral Extraction Sites	204	67
3	Non Irrigated Arable Land	614	542
4	Fruit Trees	150	79
5	Olive Groves	1768	1401
6	Conierous Forest	361	500
7	Dense Sclerophyllous	5035	3793
8	Sparse Sclerophyllous	6374	2803
9	Sparsely Vegetated Areas	1754	404
10	Rocks and Sand	492	487
11	Water	1612	1393
12	Coastal Water	398	451
	Total	20024	12208

097 unlabeled background pixels in Loukia, the IR of Dioni is 42.49, and the IR of Loukia is 56.61. In this article, Dioni is used as the training set and Loukia is used as the test set, and the number of each class in Dioni and Loukia is shown in Table V.

B. Experimental Setup

In the comparison of training and test on one image, the optimal parameters are determined on the Indian Pines dataset, and any parameters are not changed when the proposed method is applied to the Pavia University, KSC and Houston datasets to demonstrate good robustness and generalization. The side length w of the local window is 37 and the side length t of the DCT window is 5, both of which are obtained through experiments. The parameters of LightGBM are obtained by a grid search and fine-tuning, learning_rate = 0.1, boosting_type = goss, max_depth = 9, num_leaves = 10, min_data_in_leaf = 100, feature_fraction = 0.1, and num_boost_round = 1000, where feature_fraction = 0.1 indicates that a subset of 10% features is randomly sampled for

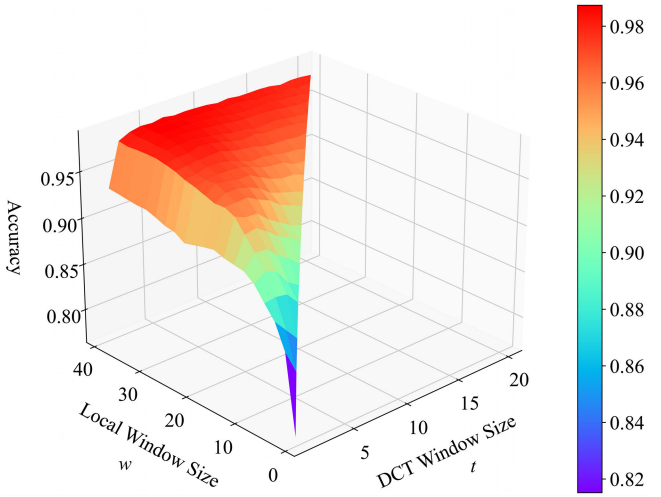


Fig. 5. Parameter selection of the local window w and the DCT window t.

training in each iteration, avoiding curse of dimensionality. In the comparison of two images separately for training and test, w and t are determined as 13 and 3 through experiments on the Dioni image of the HyRANK dataset.

The evaluation metrics for classification performance are: overall accuracy (OA), average accuracy (AA), class accuracy, and kappa coefficient. Each evaluation metric is the mean value of ten independent experiments. To evaluate the performance of the proposed method on the imbalanced datasets without changing the real class distribution, the strategy of dataset splitting selects stratified random sampling. This strategy maintains the natural imbalance of the original dataset in the training and test sets and avoids resampling the entire dataset before splitting it into a training and a test sets, which would be equivalent to resample the training and test sets as well.

C. Parameter Analysis of Local Window and DCT Window

The DCT window allows the local window to obtain larger patches, and the dual window nested structure makes it necessary for both the local window size w and the DCT window size t to be considered simultaneously during parameter tuning. The local window size w determines the patch size, and larger patches contain more background pixels. The DCT window size t determines the amount of the compressed spatial information retained and affects the dimensions of the spectral–spatial features directly. The DCT window is a feature selection process, which aims to obtain higher accuracy with the spectral–spatial features of lower dimensions, and its maximum window is set to half the size of the local window in this experiment. On the Indian Pines dataset, this experiment ranges for w from 1 to 39 in steps of 2 and for t from 1 to $((w + 1)/2)$ in steps of 1. Fig. 5 shows a 3-D plot of the parameter selection process, and the results in the plot show that the larger local window and the larger DCT window achieve higher accuracy. The purpose of this experiment is to determine the optimal window size with high-accuracy SSI features, where the local window size should be as large as possible to obtain sufficient spatial information and the DCT

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TABLE VI
COMPARISON OF DIFFERENT FEATURES ON THE INDIAN PINES DATASET

Class	SS	NDVI	NDMI	NDWI	GRVI	GNDVI	IO	Index	SSI
Alfalfa	92.93	93.90	93.41	93.17	93.90	93.41	93.41	47.32	94.39
Corn-notill	97.62	97.77	97.71	97.63	97.65	97.77	97.62	33.77	97.87
Corn-mintill	97.60	98.33	98.07	97.93	97.94	98.43	97.99	35.60	98.73
Corn	98.03	98.03	98.12	98.17	98.03	98.36	98.08	52.72	98.12
Grass-pasture	96.64	96.87	96.69	96.71	96.97	96.85	96.80	67.29	97.03
Grass-trees	98.33	99.19	98.90	98.30	99.22	99.04	99.25	66.85	99.79
Grass-pasture-mowed	84.80	86.00	86.80	85.60	84.80	86.00	84.00	23.60	86.80
Hay-windrowed	98.58	98.60	98.58	98.56	98.42	98.79	98.47	89.16	98.79
Oats	53.89	53.33	53.33	56.11	51.11	55.56	55.00	20.00	51.11
Soybean-notill	97.30	97.51	97.63	97.42	97.53	97.52	97.45	58.39	97.62
Soybean-mintill	98.73	99.00	98.95	98.64	98.99	98.81	98.97	48.66	99.22
Soybean-clean	96.67	96.97	96.72	96.87	96.61	96.85	96.80	35.47	97.21
Wheat	96.97	97.62	97.57	97.57	97.51	98.00	97.73	76.22	98.38
Woods	99.17	99.44	99.43	99.15	99.32	99.39	99.39	91.83	99.53
Buildings-Grass-Trees-Drives	97.29	98.01	98.16	97.46	98.01	97.61	97.67	32.51	98.33
Stone-Steel-Towers	85.00	87.98	87.26	86.79	86.55	87.14	85.95	88.45	90.48
OA	97.77	98.13	98.06	97.84	98.02	98.07	98.01	55.35	98.38
AA	93.10	93.66	93.58	93.50	93.28	93.72	93.41	54.24	93.96
Kappa	97.45	97.87	97.78	97.54	97.74	97.80	97.73	49.75	98.15

window should be as small as possible to achieve low feature dimensions. The highest accuracy of 98.87% is achieved with the optimal $w = 37$ for the local window and the optimal $t = 5$ for the DCT window. The side length 37 of the local window and the side length 5 of the DCT window are set as the optimal parameters of the proposed method and adopted for the Indian Pines, Pavia University, KSC, and Houston datasets. In the comparison of two images separately for training and test, the proposed method splits 50% of Dioni of the HyRANK dataset for training and the rest for test to redetermine the optimal parameters of w and t, and $w = 13$ and $t = 3$ are determined through experiments.

D. Effect of Index Features on Results

The effects of different features on results are compared on the Indian Pines dataset in Table VI, and the features for comparison include spectral–spatial features only (SS), different index features individually fused with spectral–spatial features (NDVI, NDMI, NDWI, GRVI, GNDVI, and IO), index features only (Index), and the proposed spectral–spatial index features (SSI). Table VI shows that each of the six index features fused with spectral–spatial features can improve OA, where NDVI improves OA the most and improves the accuracy of 14 classes except Corn and Oats. Only index features are difficult to reach satisfactory classification results, but fusing them with spectral–spatial features can improve OA and the accuracy of 15 classes except Oats, and the fusion of six index features with spectral–spatial features is better than the fusion of each index feature individually with spectral–spatial features. Among the 16 land cover classes of the Indian Pines dataset, the fusion of six index features with spectral–spatial features provides the largest improvement of 5.48% for Stone-Steel-Towers.

E. Effect of Sampling Strategies on Results

Table VII shows the results of no sampling strategy, random undersampling (RUS), ROS, SMOTE, combination of RUS and ROS, and combination of RUS and SMOTE on the Indian Pines dataset, and no sampling strategy is employed as Baseline. In the combination scheme, the undersampling ratio is set to half of the majority samples. As shown in Table VII, only the oversampling methods improve OA, and some of the training data are discarded in both the undersampling methods and the combination methods, which results in loss of information and reduces the classifier performance. In this experiment, the oversampling methods can better handle the binary imbalance generated by OVA, and SMOTE has a better OA than ROS by 0.11% due to synthesizing new samples, which reduces the risk of overfitting.

F. Comparison of Training and Test on One Image

Fig. 6 shows the false-color image, the corresponding ground truth, and the predicted maps of the Indian Pines dataset. The proposed method yields a smooth predicted map similar to deep learning-based methods, with less salt-and-pepper noises and a clear boundary, benefiting from the contribution of the spectral–spatial features extracted from large patches. Table VIII demonstrates the classification results of different methods, all of which split 10% for training and the rest for test. The proposed method reaches the best OA and performs best on Corn, Grass-trees, Soybean-mintill, Soybean-clean, Woods, and Buildings-Grass-Trees-Drives, which may be the optimal parameter selection performed on the Indian Pines dataset, resulting in a good fit. The proposed method performs poorly on Oats, which has only two samples in the training set. SMOTE synthesizes new samples with the nearest-neighbor parameter $K = 1$, that is, interpolation

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TABLE VII
COMPARISON OF DIFFERENT SAMPLING STRATEGIES ON THE INDIAN PINES DATASET

Class	Baseline	RUS	ROS	SMOTE	RUS + ROS	RUS + SMOTE
Alfalfa	89.51	6.59	93.66	94.39	87.32	87.56
Corn-notill	97.29	81.67	97.89	97.87	96.12	96.05
Corn-mintill	96.71	65.09	97.86	98.73	95.78	96.53
Corn	96.95	48.73	98.54	98.12	96.06	96.62
Grass-pasture	96.57	52.30	96.67	97.03	95.56	96.14
Grass-trees	98.86	81.31	99.74	99.79	98.80	98.95
Grass-pasture-mowed	84.00	32.40	86.40	86.80	83.20	82.00
Hay-windrowed	98.81	78.40	98.84	98.79	98.35	98.23
Oats	50.00	0.00	51.11	51.11	39.44	43.33
Soybean-notill	96.46	65.19	97.57	97.62	95.15	95.45
Soybean-mintill	98.81	94.36	99.21	99.22	97.83	97.66
Soybean-clean	96.16	41.52	96.74	97.21	95.17	95.22
Wheat	98.16	72.22	98.43	98.38	96.97	97.08
Woods	99.27	93.61	99.53	99.53	98.62	98.75
Buildings-Grass-Trees-Drives	97.15	67.18	98.33	98.33	96.37	96.97
Stone-Steel-Towers	88.69	15.24	91.67	90.48	85.83	85.48
OA	97.62	76.66	98.27	98.38	96.68	96.81
AA	92.71	55.99	93.89	93.96	91.04	91.38
Kappa	97.28	72.87	98.03	98.15	96.21	96.36

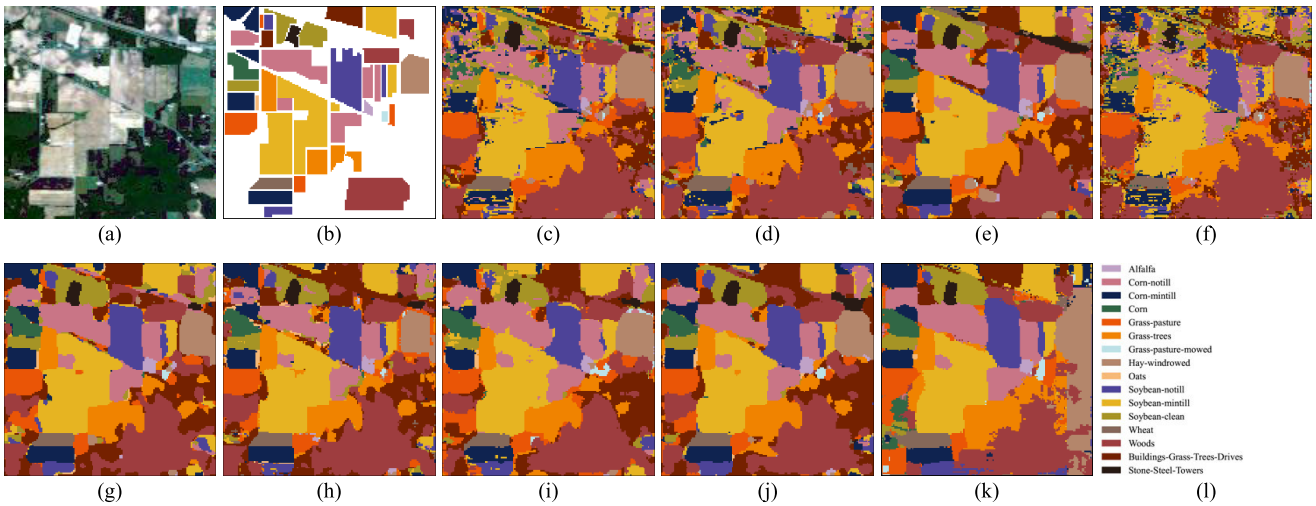


Fig. 6. Indian Pines dataset. (a) False-color image. (b) Ground truth. (c) ContextualNet. (d) SSAN. (e) SSAtt. (f) RSSAN. (g) A^2S^2K-ResNet. (h) SSSAN. (i) SSTN. (j) CVSSN. (k) TwoWin-SOVA. (l) Legend.

between the concatenation of the two samples, which limits the performance of SMOTE. For minority samples with sparse numbers, the spatial information of more other classes in the large patch may mislead the discrimination of the ensemble classifiers. The Indian Pines dataset has a high IR with a significant problem of class imbalance, and the proposed method demonstrates as good classification performance as the state-of-the-art deep learning-based methods on the imbalanced HSI dataset.

Fig. 7 shows the false-color image, the corresponding ground truth, and the predicted maps of the Pavia University dataset. The proposed method yields a smooth predicted map that has less difference from the predicted maps of the deep learning-based methods. The Pavia University image is a scene with many small patches, and the spectral–spatial information extracted from large patches predicts the land cover classes of most small patches well. Table IX demonstrates the classification results of different methods, all of which split 10% for training and the rest for test. The proposed method reaches the fourth best OA, which is 0.21% less than the best OA of CVSSN, and performs best on Painted metal sheets and Bare Soil. The Pavia University dataset has an IR of 19.69 with low class imbalance, and all of the methods achieve excellent classification performance.

Fig. 8 shows the false-color image, the corresponding ground truth, and the predicted maps of the KSC dataset. The predicted map of the proposed method is fuzzy at the edges of some patches and a little oversmoothed, which may

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TABLE VIII
COMPARISON OF DIFFERENT METHODS ON THE INDIAN PINES DATASET (10% FOR TRAINING)

Class	ContextualNet	SSAN	SSAtt	RSSAN	A^2S^2K-ResNet	SSSAN	SSTN	CVSSN	TwoWin-SOVA
Alfalfa	89.62	92.26	94.94	87.65	91.40	88.58	89.32	93.19	94.39
Corn-notill	76.24	85.69	94.89	81.85	97.88	94.05	93.09	97.30	97.87
Corn-mintill	79.53	86.35	97.04	84.33	98.44	95.80	94.68	99.18	98.73
Corn	82.71	87.33	97.49	76.79	96.69	95.01	97.36	97.33	98.12
Grass-pasture	92.44	95.05	96.71	91.13	97.56	96.33	91.65	97.63	97.03
Grass-trees	95.62	93.93	97.89	93.33	98.86	98.27	98.44	99.57	99.79
Grass-pasture-mowed	95.06	88.26	96.71	83.93	92.94	83.82	84.56	96.97	86.80
Hay-windrowed	95.70	96.69	97.87	95.03	98.74	97.60	99.51	99.56	98.79
Oats	92.88	82.84	96.12	93.65	94.33	86.24	79.88	96.66	51.11
Soybean-notill	85.84	90.16	95.33	87.40	96.63	95.69	93.47	97.67	97.62
Soybean-mintill	87.10	90.46	95.86	85.85	98.46	96.06	96.10	98.40	99.22
Soybean-clean	72.47	82.29	93.62	72.61	96.37	90.24	95.27	97.13	97.21
Wheat	98.06	97.39	99.62	94.04	99.84	99.15	99.02	99.47	98.38
Woods	95.80	96.42	98.11	94.48	98.73	98.10	97.90	98.47	99.53
Buildings-Grass-Trees-Drives	87.90	90.27	93.37	82.63	94.78	93.91	91.24	95.62	98.33
Stone-Steel-Towers	93.65	91.88	96.39	89.59	94.93	97.36	90.83	96.21	90.48
OA	86.59	90.50	96.18	86.78	97.85	95.74	95.23	98.10	98.38
AA	88.79	90.46	96.37	87.14	96.66	94.14	93.27	97.52	93.96
Kappa	84.65	89.14	95.64	84.87	97.55	95.13	94.56	97.83	98.15

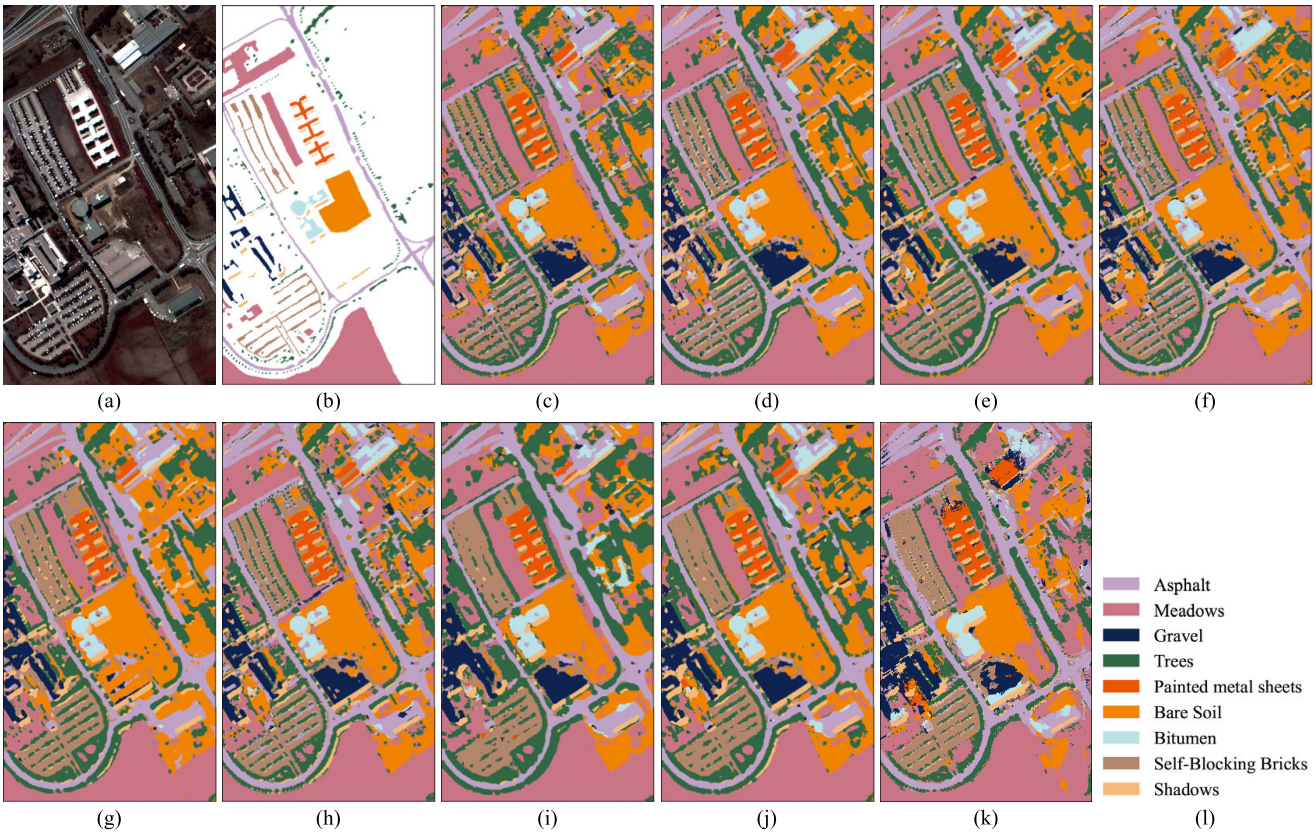


Fig. 7. Pavia University dataset. (a) False-color image. (b) Ground truth. (c) ContextualNet. (d) SSAN. (e) SSAtt. (f) RSSAN. (g) A^2S^2K-ResNet. (h) SSSAN. (i) SSTN. (j) CVSSN. (k) TwoWin-SOVA. (l) Legend.

be the limitation of the spectral–spatial features extracted from large patches. Table X demonstrates the classification results of different methods, all of which split 10% for training and the rest for test. The proposed method reaches the best OA, which is 1.06% higher than the second best CVSSN, and performs best on 11 out of 13 classes except Cattail marsh and Mud flats. The KSC dataset has an IR of 8.83 with low class imbalance, and the proposed method achieves better classification performance than the deep learning-based methods.

Fig. 9 shows the false-color image, the corresponding ground truth, and the predicted maps of the Houston dataset. The Houston image is a complex and large urban scene, and the proposed method also yields a similar predicted map as

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TABLE IX
COMPARISON OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET (10% FOR TRAINING)

Class	ContextualNet	SSAN	SSAtt	RSSAN	A^2S^2K-ResNet	SSSAN	SSTN	CVSSN	TwoWin-SOVA
Asphalt	98.78	99.33	99.39	98.46	99.92	99.79	98.52	99.89	99.65
Meadows	99.80	99.88	99.93	99.85	99.95	99.87	99.63	99.98	99.95
Gravel	97.64	98.87	99.54	97.88	99.84	99.35	98.17	99.92	98.63
Trees	99.62	99.87	99.83	99.34	99.91	99.76	98.03	99.76	99.06
Painted metal sheets	99.90	99.74	99.79	99.34	99.93	99.92	99.53	99.88	99.98
Bare Soil	99.58	99.85	99.70	99.39	99.90	99.80	99.59	99.92	99.96
Bitumen	98.16	99.30	99.57	98.29	99.98	99.67	99.67	99.91	99.48
Self-Blocking Bricks	96.46	98.15	98.27	96.87	99.27	99.05	96.94	99.57	99.26
Shadows	99.85	99.87	99.96	99.37	99.96	99.99	97.59	99.75	99.26
OA	99.16	99.57	99.63	99.11	99.87	99.74	98.97	99.90	99.69
AA	98.87	99.43	99.55	98.76	99.85	99.69	98.63	99.84	99.47
Kappa	98.89	99.43	99.51	98.83	99.83	99.66	98.64	99.86	99.59

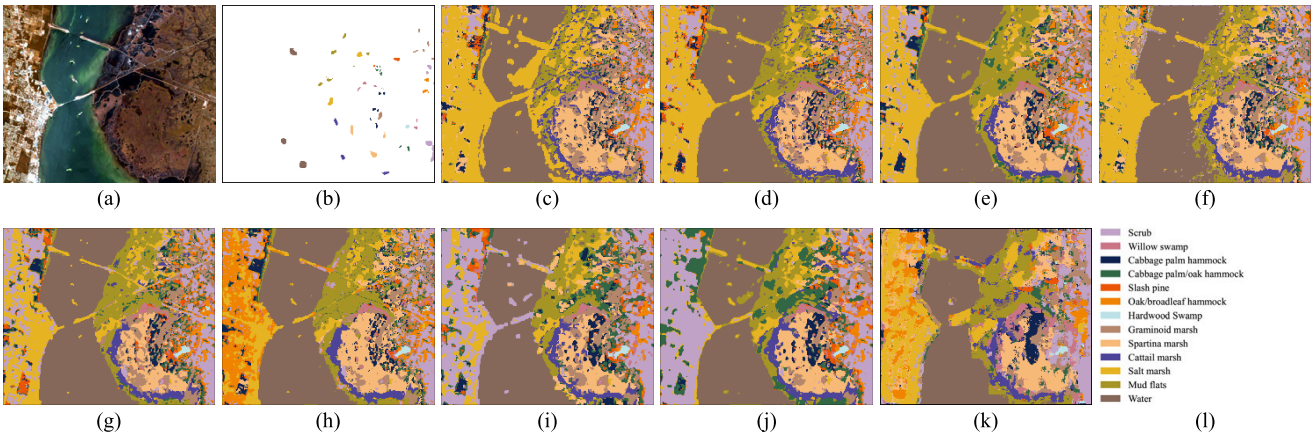


Fig. 8. KSC dataset. (a) False-color image. (b) Ground truth. (c) ContextualNet. (d) SSAN. (e) SSAtt. (f) RSSAN. (g) A^2S^2K-ResNet. (h) SSSAN. (i) SSTN. (j) CVSSN. (k) TwoWin-SOVA. (l) Legend.

TABLE X
COMPARISON OF DIFFERENT METHODS ON THE KSC DATASET (10% FOR TRAINING)

Class	ContextualNet	SSAN	SSAtt	RSSAN	A^2S^2K-ResNet	SSSAN	SSTN	CVSSN	TwoWin-SOVA
Scrub	99.16	98.70	98.99	98.78	99.08	99.38	99.35	99.77	99.91
Willow swamp	84.33	87.42	96.57	92.47	96.53	95.44	95.65	98.72	99.27
Cabbage palm hammock	76.38	75.00	88.60	78.51	89.17	91.98	92.73	96.47	99.83
Cabbage palm/oak hammock	70.75	74.05	82.15	73.94	86.47	85.32	79.94	89.93	95.77
Slash pine	63.62	69.69	81.87	69.40	88.28	78.76	77.15	95.48	98.76
Oak/broadleaf hammock	87.17	84.81	90.43	84.04	92.34	89.32	93.75	94.09	100.00
Hardwood Swamp	88.90	83.10	94.73	89.80	96.68	88.47	91.74	97.72	99.89
Graminoid marsh	92.08	93.00	97.50	95.81	97.28	95.07	97.56	99.49	99.85
Spartina marsh	96.02	95.42	99.01	96.90	99.16	99.33	98.11	99.72	100.00
Cattail marsh	93.18	92.90	99.89	94.86	99.49	99.46	98.96	100.00	99.92
Salt marsh	98.94	99.79	99.69	99.32	99.24	99.76	99.40	99.89	100.00
Mud flats	96.57	96.00	99.14	93.81	98.79	99.84	98.96	99.48	99.69
Water	99.55	99.40	99.57	98.93	100.00	99.95	99.54	100.00	100.00
OA	92.61	92.64	96.65	93.39	97.15	96.27	96.36	98.59	99.65
AA	88.20	88.41	94.47	89.74	95.58	94.01	94.07	97.75	99.45
Kappa	91.77	91.81	96.26	92.63	96.82	95.86	95.95	98.43	99.61

the deep learning-based methods. Table XI demonstrates the classification results of different methods, all of which split 10% for training and the rest for test. The proposed method reaches the OA only behind CVSSN and A^2S^2K-ResNet, which is 2.38% less than the best CVSSN, and performs best on Highway. The Houston dataset has an IR of 3.90 with low class imbalance, and the proposed method performs similar to the deep learning-based methods.

G. Comparison of Two Images Separately for Training and Test

The above four datasets are trained and tested on one image, this article explores the comparison of training and test on two separate images, and all of the methods take the Dioni image of the HyRANK dataset as the training set and the Loukia image as the test set. Fig. 10 shows the false-color

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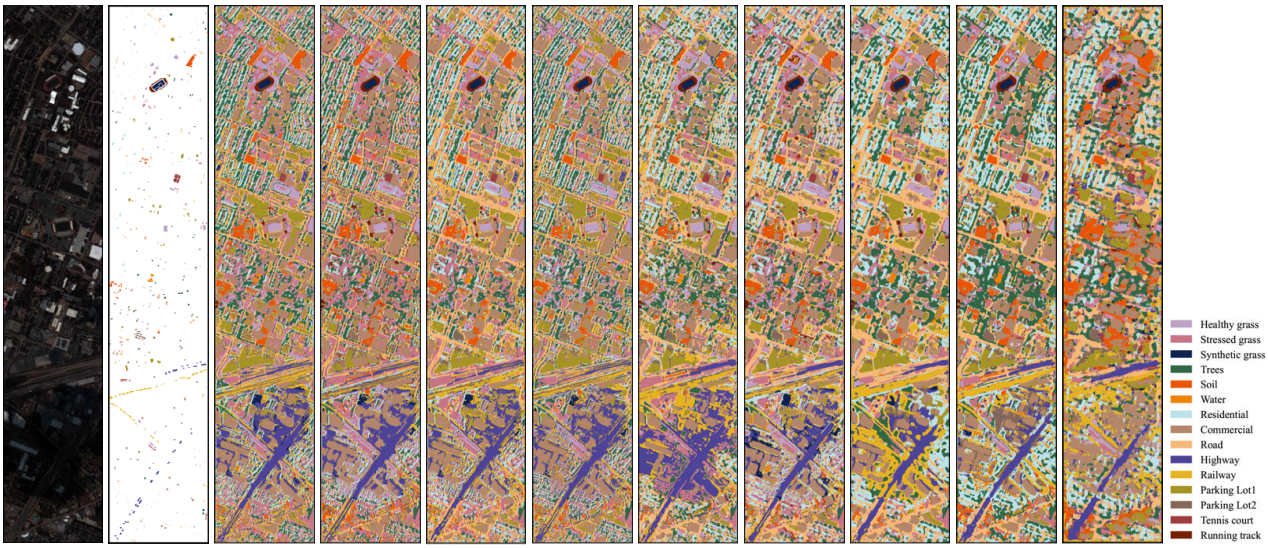


Fig. 9. Houston dataset. (a) False-color image. (b) Ground truth. (c) ContextualNet. (d) SSAN. (e) SSAtt. (f) RSSAN. (g) A^2S^2K-ResNet. (h) SSSAN. (i) SSTN. (j) CVSSN. (k) TwoWin-SOVA. (l) Legend.

TABLE XI
COMPARISON OF DIFFERENT METHODS ON THE HOUSTON DATASET (10% FOR TRAINING)

Class	ContextualNet	SSAN	SSAtt	RSSAN	A^2S^2K-ResNet	SSSAN	SSTN	CVSSN	TwoWin-SOVA
Scrub	99.16	98.70	98.99	98.78	99.08	99.38	99.35	99.77	99.91
Willow swamp	84.33	87.42	96.57	92.47	96.53	95.44	95.65	98.72	99.27
Cabbage palm hammock	76.38	75.00	88.60	78.51	89.17	91.98	92.73	96.47	99.83
Cabbage palm/oak hammock	70.75	74.05	82.15	73.94	86.47	85.32	79.94	89.93	95.77
Slash pine	63.62	69.69	81.87	69.40	88.28	78.76	77.15	95.48	98.76
Oak/broadleaf hammock	87.17	84.81	90.43	84.04	92.34	89.32	93.75	94.09	100.00
Hardwood Swamp	88.90	83.10	94.73	89.80	96.68	88.47	91.74	97.72	99.89
Graminoid marsh	92.08	93.00	97.50	95.81	97.28	95.07	97.56	99.49	99.85
Spartina marsh	96.02	95.42	99.01	96.90	99.16	99.33	98.11	99.72	100.00
Cattail marsh	93.18	92.90	99.89	94.86	99.49	99.46	98.96	100.00	99.92
Salt marsh	98.94	99.79	99.69	99.32	99.24	99.76	99.40	99.89	100.00
Mud flats	96.57	96.00	99.14	93.81	98.79	99.84	98.96	99.48	99.69
Water	99.55	99.40	99.57	98.93	100.00	99.95	99.54	100.00	100.00
OA	92.61	92.64	96.65	93.39	97.15	96.27	96.36	98.59	99.65
AA	88.20	88.41	94.47	89.74	95.58	94.01	94.07	97.75	99.45
Kappa	91.77	91.81	96.26	92.63	96.82	95.86	95.95	98.43	99.61

TABLE XII
COMPARISON OF DIFFERENT METHODS ON THE HYRANK DATASET (10% FOR TRAINING)

Class	ContextualNet	SSAN	SSAtt	RSSAN	A^2S^2K-ResNet	SSSAN	SSTN	CVSSN	TwoWin-SOVA
Dense Urban Fabric	8.57	11.38	7.27	12.70	5.98	10.44	19.22	4.84	10.07
Mineral Extraction Sites	7.53	8.72	0.00	8.12	7.67	0.30	0.00	7.23	0.00
Non Irrigated Arable Land	10.00	41.10	25.25	47.67	34.00	32.92	31.09	27.12	43.36
Fruit Trees	8.70	47.56	9.59	15.32	18.28	12.42	4.38	6.78	31.65
Olive Groves	54.31	84.76	48.83	59.69	100.00	45.50	68.94	34.67	0.64
Conierous Forest	76.92	3.12	40.70	28.93	0.00	70.39	100.00	13.41	10.20
Dense Sclerophyllous Vegetation	71.96	66.69	76.57	75.91	75.56	69.57	71.92	70.63	84.18
Sparse Sclerophyllous Vegetation	27.05	36.92	28.89	38.60	33.72	32.04	40.11	46.28	22.05
Sparsely Vegetated Areas	11.63	17.68	11.43	15.28	23.69	13.10	12.62	1.97	66.09
Rocks and Sand	0.40	0.00	33.33	37.22	32.54	29.85	23.06	20.30	62.83
Water	100.00	93.49	99.29	98.93	99.93	100.00	100.00	100.00	100.00
Coastal Water	98.46	61.12	96.16	98.60	100.00	96.00	74.92	77.56	100.00
OA	38.16	46.39	40.27	48.39	41.40	42.96	45.45	39.67	53.87
AA	39.63	39.38	39.78	44.75	44.28	42.71	45.52	34.23	44.26
Kappa	28.74	35.89	32.62	40.35	32.28	33.92	36.70	31.70	42.80

image, the corresponding ground truth, and the predicted maps of the HyRANK dataset, and the predicted maps on this extremely challenging dataset differ significantly from each other. Table XII demonstrates the classification results of

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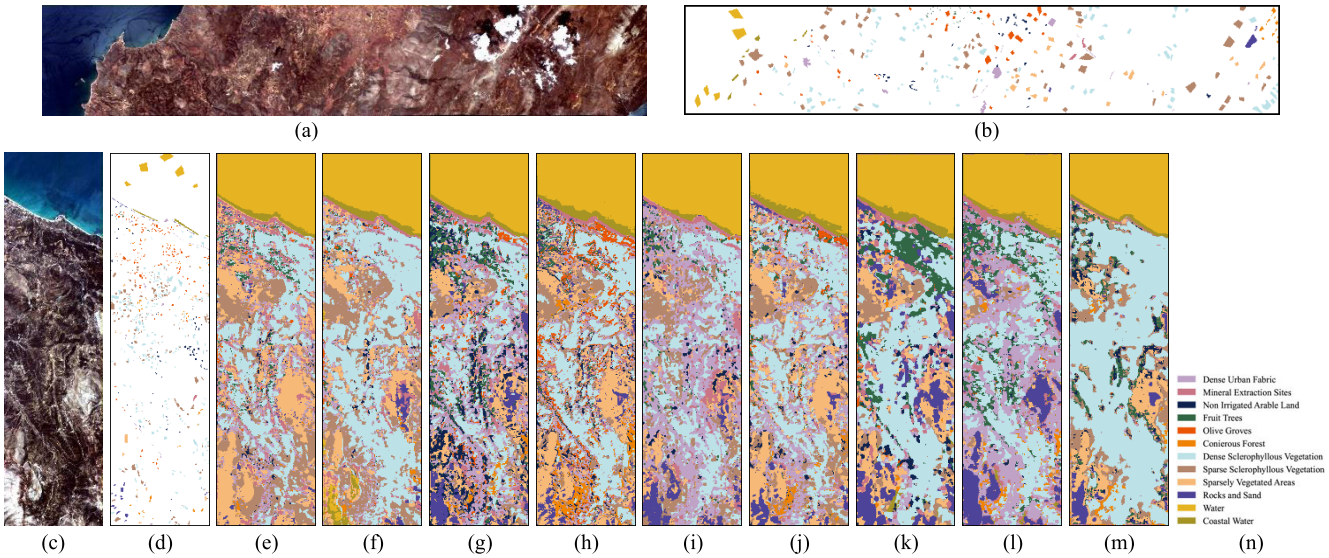


Fig. 10. HyRANK dataset. (a) False-color image of Dioni. (b) Ground truth of Dioni. (c) False-color image of Loukia. (d) Ground truth of Loukia. (e) ContextualNet. (f) SSAN. (g) SSAtt. (h) RSSAN. (i) A^2S^2K-ResNet. (j) SSSAN. (k) SSTN. (l) CVSSN. (m) TwoWin-SOVA. (n) Legend.

different methods, and the proposed method reaches the best OA of 53.87%, which is 5.48% higher than the second best RSSAN. The Dioni dataset has an IR of 42.49 and the Loukia dataset has an IR of 56.61, both of which have significant class imbalance, and the proposed method still handles the problem of class imbalance well in the condition that the two images are separately used for training and test and achieves classification performance superior to the deep learning-based methods.

V. Discussion

A. SHAP Explanation of the Contribution of the SSI Features to the Prediction Results

SHAP is a tool for interpretability to understand the process of a model of machine learning using input features to make predictions through SHAP values [48], [49], [50]. This article uses TreeExplainer, which is a local explanation method for tree models in SHAP, to obtain force plots for explaining the contributions of the extracted SSI features to the predictions. Taking the first sample of the Indian Pines test set as an example, Fig. 11 shows the contributions of the SSI features to the prediction results in the 16 base classifiers of the SOVA ensemble classifiers. The 16 base classifiers correspond to the 16 land cover classes in the Indian Pines dataset, and each base classifier outputs the prediction value of the sample belonging to this class or not. In Fig. 11, red represents that input features push prediction values higher and blue represents that input features push prediction values lower. The given sample belongs to the class 14 of the Indian Pines dataset. The red part of the base classifier 14 in Fig. 11 is much longer than the red parts of the other base classifiers, indicating that the SSI features contribute significantly to the accurate discrimination of the input test sample by the base classifiers. The blue part of the base classifier 14 in Fig. 11 is much shorter than the blue parts of the other base classifiers, indicating that the SSI features effectively suppress

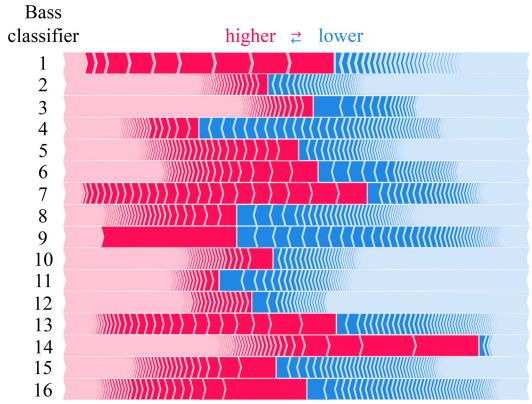


Fig. 11. Force plots of sample 1 in the Indian Pines test set.

the other base classifiers from discriminating the input test sample into their corresponding classes, avoiding incorrect predictions. The more discriminative SSI features extracted by the proposed method can provide great contributions to high-accuracy prediction.

The dependence plot of SHAP shows the interaction effect between Feature-426 and NDVI on the prediction results of each sample in the Indian Pines test set. As shown in Fig. 12, this article takes NDVI in the index features as an example and outputs Feature-426 in the spectral–spatial features using approximate_interactions in SHAP, which has the highest possibility of interaction with NDVI. The X-axis is the normalized feature value, the Y-axis is the SHAP value of the feature, and the color legend is the degree of the interaction effect between NDVI and Feature-426, with red representing a high degree and blue a low degree. The coloring pattern in Fig. 12 indicates the interaction between NDVI and Feature-426, and the combination of two highly correlated features is better than independently [51]. The spectral–spatial features and the index features fused by the proposed method interact to make a contribution to the predictions.

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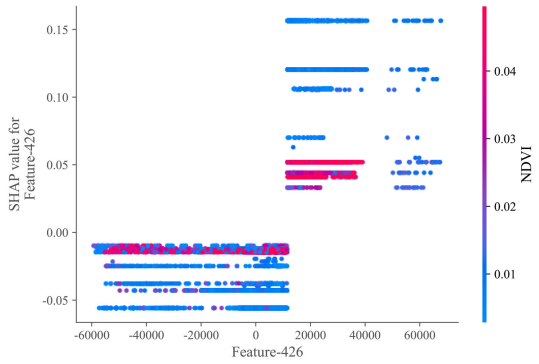


Fig. 12. Dependence plot between feature 426 and NDVI in the Indian Pines test set.

TABLE XIII
RUNNING TIME (S) OF EACH MODULE OF THE PROPOSED TWOWIN-SOVA ON THE FIVE DATASETS

Module	TwoWin-DCT	SOVA	Others	Total
Indian Pines	60	27	1	88
Pavia University	130	20	3	153
KSC	44	13	11	68
Houston	105	29	22	156
HyRANK	107	49	6	162

B. Effect of SOVA on Results

The results of the five datasets show that the proposed method has the good OA and the relatively inferior AA. The large patches obtained by the proposed method contribute more to the predictions of the land cover classes with large areas, which means that the accuracy of the minority classes is hard to reach the high accuracy of the majority classes, resulting in the lower AA. In this article, the class distributions of the training and test sets are different, and the SOVA uses the base classifiers learned from the training set with the balanced class distribution after SMOTE to predict the samples of the test set with the imbalanced class distribution, achieving the good class accuracy and OA, which can tackle the dataset shift problem with different distributions of training and test data. Small and large patches are obtained through two different local windows, and weighting the spatial features extracted from small and large patches based on prior information such as the class distribution of the dataset is one of the possible future works in this study.

C. Analysis of Experimental Conditions and Running Time

The proposed method uses tree models to achieve the same great accuracy as deep learning, demonstrating the performance of tree models in HSI classification. All experiments are carried out with Intel Core i9-13900HX CPU and 32-GB RAM and are implemented by Python. Compared with deep learning, tree models have lower hardware requirements of experimental conditions and lower deployment costs. The running time of each module for the five datasets in Table XIII shows that the time cost of the proposed method concentrates in the module of TwoWin-DCT. The TwoWin-DCT takes the patches of each spectral band for each pixel as the DCT inputs. The compression process of DCT requires substantial computation time, and the running time of this process is related to the number of labeled pixels in the dataset, and more labeled pixels lead to more computation time. The SOVA uses LightGBM as the base classifier, which can learn rapidly from high-dimensional large samples and reduce the time cost. Parallelizing the process of DCT can reduce the cost of computation and is a future work of this study.

VI. CONCLUSION

In this article, a TwoWin-SOVA is proposed to address the problems of high dimensionality caused by large patches and class imbalance in HSI classification. The TwoWin-DCT is used for feature extraction, retaining the information clustered in the upper left corner of the larger patches to extract the spectral–spatial features. The index features represent the correlation between spectral bands and are fused with the spectral–spatial features into the SSI features. The SOVA constructs ensemble classifiers by the OVA strategy with SMOTE, in which the base classifier is LightGBM, achieving fast and high-accuracy classification. The SHAP explanations illustrate that the SSI features extracted by the TwoWin-SOVA make great contributions to the prediction results of the base classifiers in the SOVA. The TwoWin-SOVA validates similar classification performance with the state-of-the-art deep learning-based methods on the five public HSI datasets of the Indian Pines, Pavia University, KSC, Houston, and HyRANK, effectively dealing with the problems of high dimensionality caused by large patches and class imbalance. Compared with deep learning, tree models in the TwoWin-SOVA have lower cost of deployment and wider application scenarios.

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